

Customer Purchase Behavior Analysis

Unlocking retail insights through Python, SQL, and Power BI





The Business Challenge

Fluctuating Patterns

Purchase behavior varies across demographics, products, and channels.

Key Questions

What drives customer decisions?
How do discounts, ratings, and payment methods influence purchases?

Strategic Goal

Enhance marketing, optimize pricing, and improve retention through data-driven insights.

Methodology: Three-Stage Approach



Python Prep

SQL Analysis

Power BI

Stage 1: Python Data Preparation

01

Load & Explore

3,900 transactions analyzed and cleaned.

02

Handle Missing Data

Imputed median review ratings by category.

03

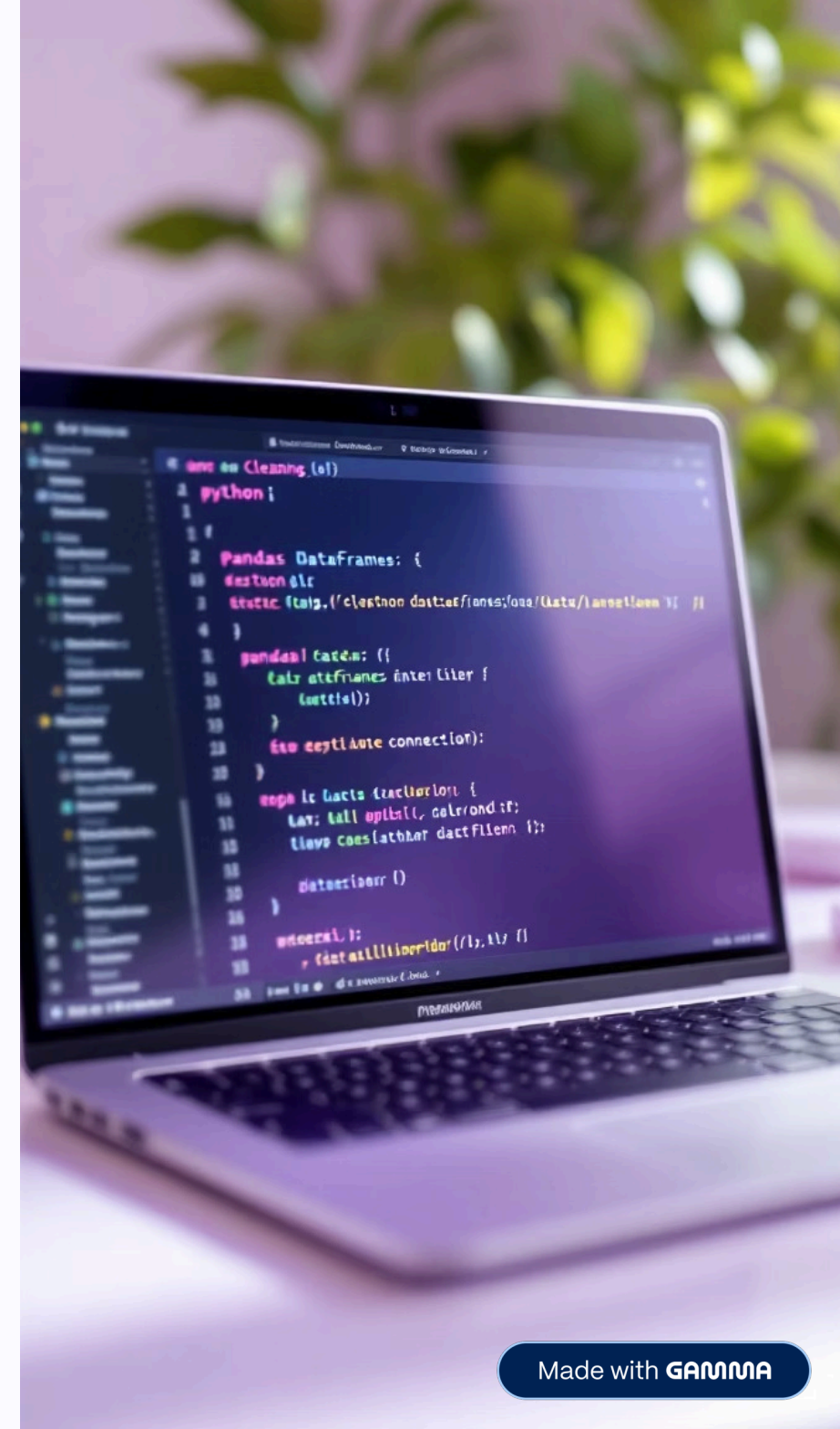
Feature Engineering

Created age_group and purchase_frequency_days columns.

04

Database Integration

Connected to PostgreSQL via SQLAlchemy and Psycopg2.



Stage 2: SQL Analysis & Insights

Revenue Segmentation

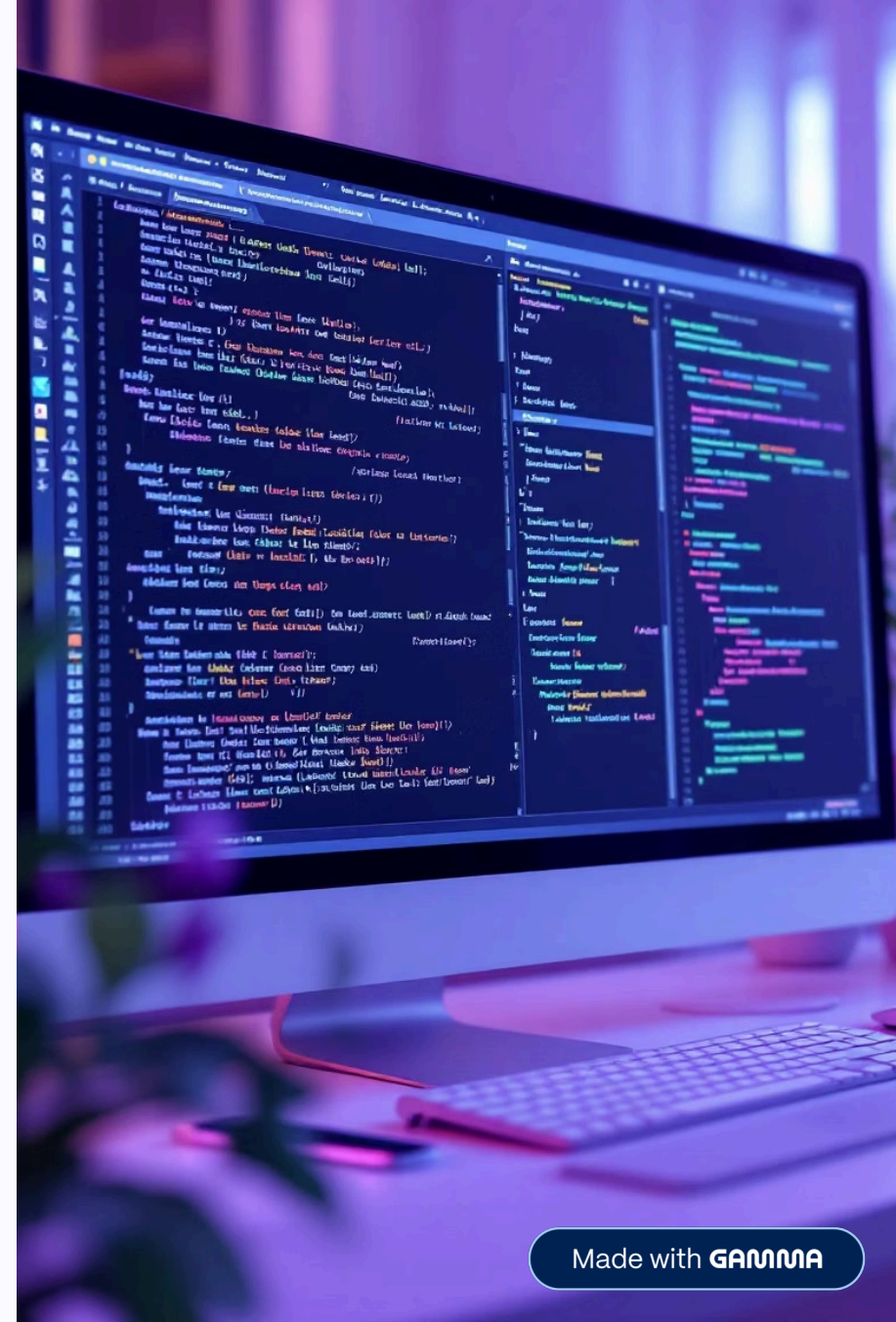
Analyzed by gender, subscription status, age group.

Product Performance

Identified top performers and discount effectiveness.

Customer Loyalty

Segmented by purchase frequency and repeat behavior.



Key Findings



Discount Impact

Customers using discounts showed above-average purchase amounts.



Subscription Value

Subscribers contributed higher average spending and revenue.



Premium Shipping

Express shipping correlated with higher customer spending.



Customer Loyalty

Returning customers formed majority, showing strong brand trust.



Strategic Recommendations

1 Target High-Value Segments

Focus marketing on Power BI-identified segments.

2 Boost Subscriptions

Offer loyalty rewards and exclusive discounts.

3 Optimize Inventory

Stock top-performing products to prevent lost sales.

4 Refine Discounts

Improve conversion while maintaining profit margins.

5 Monitor Continuously

Use Power BI dashboard for real-time decision-making.



Future Vision & Connect

Next Steps

- Predictive ML models for churn forecasting
- Real-time Power BI integration
- ETL pipeline automation
- Customer feedback analysis

Get in Touch

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